METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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EQUIPMENT LIST

R-201 quench converter and synthesis loop · 7 items · with design basis

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1 Purpose and scope

This list gives every equipment tag drawn on the EPC P&IDs MEOH-PID-001 and MEOH-PID-002 (R-201, E-201, E-202, E-203, E-204, V-201, C-201) with its type, main size, operating and design conditions, pressure-part material and datasheet number. Operating values are the simulator model at its design point (noise off); the maximum operating temperatures come from model upset runs; R-201 diameter and height are those of the Design tab conceptual sizing; tags and line numbers are those of the P&IDs. Design rules, allowable stresses, service U values, tube geometry and Nelson-curve limits are typical practice and are marked assumed. There is no vendor or licensor input.

2 Equipment list

Tag	Description	Type	Main size	Operating P / T	Design P / T	Material	Datasheet
R-201	Methanol quench converter	Vertical 4-bed adiabatic quench reactor, hot wall	ID 4810 × 16620 mm T/T; catalyst 184 m³	79.7 bar g / 220–277 °C	100 bar g / 335 °C	2¼Cr-1Mo	MEOH-V1-ME-DS-001
E-201	Feed/effluent interchanger (hot)	Shell and tube, vertical, single pass counter-current, HP closures	2330 m²; shell ID 1500 mm	T 77.4 / S 80 bar g; T 268→206 / S 97→254 °C	100 bar g / 335 °C	1¼Cr-½Mo	MEOH-V1-ME-DS-003
E-202	Feed/effluent interchanger (cold)	Shell and tube, horizontal, U-tube, HP closures	930 m²; shell ID 1350 mm	T 76 / S 80.6 bar g; T 206→156 / S 49→97 °C	100 bar g / 280 °C	1¼Cr-½Mo / CS / 304L	MEOH-V1-ME-DS-004
E-203	Final cooler (cooling water)	Shell and tube, horizontal, 2 shells in series	6910 m² total; shell ID 1850 mm	T 75.5 bar g, 156→40 °C; CW 32→42 °C	100 bar g / 225 °C; CW side 10 bar g / 225 °C	1¼Cr-½Mo / duplex / CS	MEOH-V1-ME-DS-005
E-204	Start-up heater (HP steam)	Shell and tube, horizontal, U-tube; steam in shell	760 m²; shell ID 1250 mm	T 79.9 bar g; S steam 40 bar g / 252 °C (start-up only)	100 bar g / 335 °C; steam side 45 bar g / 280 °C	1¼Cr-½Mo / CS	MEOH-V1-ME-DS-006
V-201	HP separator	Vertical vessel with mesh demister	ID 4200 × 5500 mm T/T	75 bar g / 40 °C	88 bar g / 120 °C	CS SA-516-70N	MEOH-V1-ME-DS-002
C-201	Loop circulator	Single-stage barrel centrifugal, variable-speed motor	1928 kNm³/h (30051 m³/h actual); 6.53 MW at 80 % speed	75 → 80.6 bar g; 40 → 49 °C	100 bar g / 120 °C	Forged CS barrel	MEOH-V1-ME-DS-007

Operating P/T: tube side (T) and shell side (S) for exchangers. Design P is in bar g, design T in °C. All items are protected by PSV-201 (set 88 bar g) on V-201, see MEOH-V1-PE-RLB-001.

3 Design basis

3.1 Design pressure — two zones protected by one PSV

PSV-201 on V-201 (set 88 bar g, MEOH-PID-002) is the only relief device of the loop. Every item it protects must have a design pressure (MAWP) at least equal to the set pressure, and during relief each item may reach 110 % of its own MAWP (ASME VIII Div. 1 UG-125). V-201 is at the low-pressure end; with V-201 at 110 % of the set and C-201 at 100 % speed (model K.F100) the discharge side is 11.3 bar higher, which sets zone B.

Zone	Equipment	Maximum operating (basis)	Rule (assumed)	Design P
Zone A — low-pressure side	V-201, C-201 suction, E-203 outlet piping	PAH 78 bar g (PIC-203 high alarm); PSHH-203 85 bar g	max(1.10 × 78, 78 + 1.7) = 85.8 bar g, and ≥ PSHH-203 + 3 bar = 88 bar g → design = PSV-201 set	**88 bar g**
Zone B — C-201 discharge to V-201 inlet	C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side	C-201 discharge 80.6 bar g at design; 108.1 bar g with V-201 relieving (110 % of set) and C-201 at 100 % speed (ΔP 11.3 bar)	ASME UG-125: every item protected by PSV-201 ≤ 110 % of its MAWP → MAWP ≥ (1.1 × 88 + 11.3) / 1.1 = 98.2 bar g, rounded up to 5 bar	**100 bar g**

3.2 Design temperature — maximum operating + 30 °C, rounded up to 5 °C (assumed)

Maximum operating temperatures are the highest values seen in six model upset runs of 60 min (trips as installed; values recorded only while the loop circulates at ≥ 30 % of design): Instrument-air failure (all valves to their fail positions); Loss of most E-203 cooling water (75 % less flow; design case); C-201 trips (motor), make-up on until FSLL-201; FV-202 fails wide open (FIC-202 MAN 100 %); TV-211 E-201 bypass to 100 % (TIC-211 MAN); Hot afternoon: cooling water 38 °C. For R-201, E-201 and E-204 the TSHH-2xx trip setting is used when higher.

Item	Normal (model)	Max operating	Upset run	Design T
R-201 shell (hot wall, bed temperature)	277 °C	302 °C	TV-211 E-201 bypass to 100 % (TIC-211 MAN)	**335 °C**
E-201 tube side (converter effluent in)	268 °C	302 °C	TV-211 E-201 bypass to 100 % (TIC-211 MAN)	**335 °C**
E-201 shell side (bed-1 feed out)	254 °C	290 °C	TV-211 E-201 bypass to 100 % (TIC-211 MAN) (raised to the tube-side design: one exchanger, one design T)	**335 °C**
E-202 tube side (effluent in)	206 °C	249 °C	TV-211 E-201 bypass to 100 % (TIC-211 MAN)	**280 °C**
E-202 shell side (converter feed / quench out)	97 °C	136 °C	Loss of most E-203 cooling water (75 % less flow; design case)	**170 °C**
E-203 tube side (effluent in)	156 °C	194 °C	Loss of most E-203 cooling water (75 % less flow; design case)	**225 °C**
V-201 and C-201 suction (separator gas)	40 °C	89 °C	Loss of most E-203 cooling water (75 % less flow; design case)	**120 °C**
C-201 casing (discharge)	49 °C	90 °C	Loss of most E-203 cooling water (75 % less flow; design case)	**120 °C**
E-204 tube side (bed-1 feed, start-up heater)	220 °C	305 °C	R-201 inlet: takes the R-201 design temperature	**335 °C**

Not covered: total loss of cooling water. In the model the TSHH trip closes the make-up at 488 s but C-201 keeps circulating and the separator gas reaches about 351 °C (EQ-em-04).

4 Materials for hydrogen service

High-temperature hydrogen attack (HTHA) is avoided by keeping each steel below its API 941 Nelson curve at the design temperature and the hydrogen partial pressure. At the loop's hydrogen partial pressure (≈ 80.8 bar a at the zone B design pressure) the carbon-steel curve is near 220 °C (approximate reading, typical practice; the 2016 edition adds a lower curve for non-PWHT welded carbon steel); 1¼Cr-½Mo and 2¼Cr-1Mo are inside their curves with wide margins at ≤ 335 °C.

Item	Design T	pH_2 (bar a)	CS limit	Material	Reason
R-201	335 °C	80.8	220 °C — CS out	2¼Cr-1Mo SA-387 Gr 22 Cl 2; forgings SA-336 F22 Cl 3; internals SS 321	Hot wall at up to 335 °C with H_2 at 81 bar a: well above the carbon-steel Nelson curve, so a Cr-Mo steel. 2¼Cr-1Mo is the usual converter steel (1¼Cr-½Mo would also sit inside its curve); specify step-cooling (temper embrittlement) tests and PWHT.
E-201	335 °C	80.8	220 °C — CS out	1¼Cr-½Mo SA-387 Gr 11 Cl 2 shell and channel; tubes SA-213 T11	Both sides hot hydrogen-rich gas (335 °C design): carbon steel is outside its Nelson curve; 1¼Cr-½Mo is inside with a wide margin.
E-202	280 °C	77.7	220 °C — CS out	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel (effluent); shell SA-516 Gr 70 (feed side 170 °C); tubes 304L	Effluent side 280 °C is above the carbon-steel limit; the feed side (170 °C) can be carbon steel. Tubes 304L because crude methanol and wet CO_2 start to condense in this exchanger (carbonic-acid corrosion), typical practice.
E-203	225 °C	77.7	220 °C — CS out	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel; tubes duplex 2205; shell (CW) SA-516 Gr 70	Process inlet design 225 °C is just above the carbon-steel curve (≈ 220 °C), so the channel is 1¼Cr-½Mo; duplex tubes resist wet CO_2/methanol inside and chlorides in the cooling water outside.
V-201	120 °C	72.7	220 °C — CS ok	Carbon steel SA-516 Gr 70 (N), PWHT, impact tested; demister SS 316	120 °C is far below the carbon-steel Nelson curve; CA 3 mm for wet CO_2 / crude methanol. Thick wall (145 mm): PWHT and Charpy tests required.
C-201	120 °C	80.8	220 °C — CS ok	Forged carbon / low-alloy steel barrel casing (vendor standard)	Discharge design 120 °C: carbon steel is inside its Nelson curve. Impeller and shaft per vendor.

Item	Design T	pH_2 (bar a)	CS limit	Material	Reason
E-204	335 °C	80.8	220 °C — CS out	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel and tubes (process); shell SA-516 Gr 70 (steam)	Process side carries hot loop gas to R-201 (335 °C design): Cr-Mo. The steam shell (280 °C) is carbon steel.

5 Wall thickness summary (ASME VIII Div. 1, UG-27(c)(1) and UG-32(d))

t = P R / (S E − 0.6 P) for the cylinder (R inside radius, corroded) and t = P D / (2 S E − 0.2 P) for a 2:1 ellipsoidal head; corrosion allowance added and the plate rounded up. S from ASME II-D at the design temperature (typical values, assumed): SA-516 Gr 70 138 MPa; 1¼Cr-½Mo / 2¼Cr-1Mo Cl 2 147 MPa. E = 1.0 (full radiography).

Item	P (bar g)	D (mm)	S (MPa)	E	CA	Shell t req. + CA	Plate	Head t req. + CA	Head	MAWP (bar g)	Hydrotest
R-201 shell and heads	100	4810	147	1	3	170.8 + 3	**175 mm**	164.9 + 3	**170 mm**	100.7	130.9
V-201 shell and heads	88	4200	138	1	3	139.4 + 3	**145 mm**	135.0 + 3	**140 mm**	89.6	116.4

Exchanger shell and channel thicknesses are on their datasheets. R-201 at 175 mm is heavy for Div. 1; a Div. 2 design (higher allowable stress) would be thinner — a vendor option.

6 Exchangers and rotating equipment — main data

Tag	Duty (MW)	LMTD (K)	F	Model / required UA (kW/K)	Service U (W/m²K)	Area (m²)	Note
E-201	44.2	46.4	1	952	450	2330	single tube pass, true counter-current: F = 1 (the cold outlet is above the hot outlet — a temperature cross a 1-2 shell cannot take)
E-202	36.5	108.5	0.965	336	400	930	1 shell, 2 tube passes
E-203	132.9	39.9	0.966	3451	550	6910	2 shells in series, 2 tube passes each (one 1-2 shell would give F ≈ 0.78, below the usual 0.8 minimum)
E-204	22.2	64.5	1	345	500	760	condensing steam on the shell side: F = 1

C-201: 1928 kNm³/h (30051 m³/h actual), suction 75 bar g / 40 °C, discharge 80.6 bar g / 48.6 °C, pressure ratio 1.074, shaft power 6.53 MW at 80 % speed (model); 2410 kNm³/h and ≈ 12.6 MW at 100 % speed. Motor rating per the electrical load list MEOH-V1-EE-LL-001.

6.1 Loop gas volume from this list vs the model

R-201 (less catalyst solids)	V-201 (less liquid at NLL)	E-201	E-202	E-203 tubes	E-204 tubes	Piping (assumed)	Total	Model K.Vloop
216	67	18	7	26	3	40	**376**	352

m³; exchanger volumes from typical volume per m² of area for 19.05 × 2.11 mm tubes (assumed). Agreement within ± 25 % is the check.

7 Checks at issue

Check	This document	Reference	Result
R-201 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓
E-201 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓
E-202 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓
E-203 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓
E-204 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓

Check	This document	Reference	Result
V-201 design pressure ≥ PSV-201 set	88 bar g	88 bar g (MEOH-PID-002)	✓
C-201 design pressure ≥ PSV-201 set	100 bar g	88 bar g (MEOH-PID-002)	✓
Zone B ≤ 110 % of MAWP while V-201 relieves	108.1 bar g	1.1 × 100 = 110.0 bar g	✓
V-201 design ≥ PSHH-203 + 3 bar and ≥ 1.1 × PAH	88 bar g	85 + 3 = 88; 1.1 × 78 = 85.8	✓
R-201 design T ≥ TSHH + 25 °C	335 °C	300 + 25 = 325 °C	✓
R-201 ID / T/T = Design tab	4810 mm / 16.62 m	4814 mm / 16.62 m	✓
R-201 catalyst = P&ID and model	184.1 m³ / 208.3 t	184 m³ (P&ID), ΣK.Wb 208.25 t	✓
R-201 inside its Nelson curve	335 °C	≈ 510 °C (approx.)	✓
E-201 inside its Nelson curve	335 °C	≈ 480 °C (approx.)	✓
E-202 inside its Nelson curve	280 °C	≈ 480 °C (approx.)	✓
E-203 inside its Nelson curve	225 °C	≈ 480 °C (approx.)	✓
V-201 inside its Nelson curve	120 °C	≈ 220 °C (approx.)	✓
C-201 inside its Nelson curve	120 °C	≈ 220 °C (approx.)	✓
E-204 inside its Nelson curve	335 °C	≈ 480 °C (approx.)	✓
Loop volume vs K.Vloop (± 25 %)	376 m³	352 m³	✓

The Mechanical tab recomputes these checks (and more) from the live model each time it is opened.

8 Engineering queries

No.	From → to	Issue	Proposed
EQ-em-01	MEOH-V1-ME-EQL-001 → MEOH-PID-001 / MEOH-PID-002	The P&IDs give no design pressure or temperature for any equipment (V-201 shows only its operating 75 bar g / 40 °C). The equipment list sets two pressure zones: zone A (V-201, C-201 suction) 88 bar g = PSV-201 set, and zone B (C-201 discharge through R-201 and the exchangers to the V-201 inlet) 100 bar g, because with V-201 relieving at 110 % of set and C-201 at 100 % speed the discharge side is 11.3 bar higher. This also answers EQ-ep-03 (≈ 93 bar g before accumulation).	Add design P/T to every equipment tag on the next P&ID revision (R-201 100 bar g / 335 °C, V-201 88 bar g / 120 °C, …) and a spec break at the C-201 suction (zone A/B); confirm no block valves between C-201 discharge and V-201 (single PSV-201 protects both zones).
EQ-em-02	MEOH-V1-ME-DS-005 (E-203) → Simulator model / Design tab / MEOH-V1-PE-HMB-001	The model's final cooler is sensible-only (83.8 MW, CW 7740 t/h). With the condensing duty the real E-203 duty is 132.9 MW; at the model's CW flow the CW return would be about 46.8 °C, above the usual 45 °C limit for cooling-water return.	Datasheet E-203 is sized on the total 132.9 MW with 11433 t/h of CW (32 → 42 °C, assumed). Use this CW flow in the utility summary; if the model is revised, add the latent heat to the E-203 balance.
EQ-em-03	MEOH-V1-ME-EQL-001 → MEOH-V1-PE-LSL-001	The line sizing list checks all loop pipe walls at 88 bar g with A106 Gr B at 260 °C and Class 600. Zone B lines (DN750-PG-202, DN750-PG-201, DN600-PG-213, DN900-PG-251, quench lines) are 100 bar g; DN600-PG-213 and DN900-PG-251 run up to 335 °C in hydrogen service (carbon steel outside the Nelson curve).	Re-check zone B wall thickness at 100 bar g and the design temperatures of this EQL; Cr-Mo (1¼Cr-½Mo or 2¼Cr-1Mo) for the converter inlet/outlet piping and Class 900 flanges (assumed) on zone B.
EQ-em-04	MEOH-V1-ME-DS-002 / DS-007 → C&E MEOH-V1-IN-CE-001 / HAZOP MEOH-V1-SA-HZP-001	Total loss of cooling water to E-203 (model, trips as installed): TSHH-2xx any bed thermocouple above 300 °C trips the make-up at 488 s, but C-201 keeps circulating and the separator gas reaches about 351 °C (above 60 °C after 12 s); V-201 (120 °C) and C-201 (120 °C) are not designed for this.	Add a C-201 stop on high separator / C-201 suction temperature (e.g. TSHH-253 at 65 °C, assumed) to the C&E, or design V-201 and C-201 for the upset; review in the HAZOP follow-up.
EQ-em-05	MEOH-V1-ME-EQL-001 → MEOH-V1-PE-RLB-001 (HV-205 depressuring)	The model depressures isothermally (separator temperature follows the cooling water), so it cannot give the minimum metal temperature reached when HV-205 blows the loop down. V-201 and E-203 are thick carbon steel.	Run a dynamic blowdown study (adiabatic gas expansion, wall heat transfer) and set the MDMT; −29 °C with impact testing is assumed on the datasheets until then.

9 Assumptions

Design pressure rule: max(1.10 × max operating, max operating + 1.7 bar), and ≥ trip + 3 bar; zone B from ASME UG-125 accumulation. Design temperature: max operating + 30 °C, rounded up to 5 °C. Allowable stresses and Nelson limits: typical ASME II-D / API 941 readings (verify). Corrosion allowance 3 mm. Joint efficiency 1.0. MDMT −29 °C pending a blowdown study. Exchanger service U: E-201 450, E-202 400, E-203 550, E-204 500 W/m²K; installed area = model UA + 10 %. Tubes 19.05 × 2.11 mm on 23.8 mm triangular pitch. V-201: Souders–Brown K 0.077 m/s; level span = model K.Vboot. E-203 CW 32 → 42 °C on the total duty (sensible from the model + condensing from the Design tab latent heats). Weights: pressure parts + 10 %, R-201 internals 30 t and inert balls 15 t. Flange classes per ASME B16.5 (assumed).